



Chrysoberyl

△ Cat's-eye chrysoberyl cabochon

Although crystals of chrysoberyl are not uncommon, the gemstone variety alexandrite is one of the rarest and most expensive gems in the world, with specimens seldom exceeding 10 carats. Alexandrite has the extraordinary visual property of appearing green in daylight but red under tungsten light; other forms of chrysoberyl occur in green, greenish yellow, and yellow. Alexandrite was discovered in the Ural Mountains in 1830, and was named after the Russian ruler Alexander II, on whose birthday it was supposedly found.

Specification

Chemical name Beryllium aluminium oxide | **Formula** BeAl_2O_4
Colours Green, yellow, brown | **Structure** Orthorhombic
Hardness 8.5 | **SG** 3.7 | **RI** 1.74–1.76 | **Lustre** Vitreous
Streak White | **Locations** Russia, Myanmar, Zimbabwe, Tanzania, Madagascar, USA, Brazil, Sri Lanka

Rough



Gemmy crystals | These large, wedge-shaped crystals are typical of chrysoberyl and are a good colour. They display cyclic twinning (a group of crystals that have formed radially).

Colours



Cat's-eye chrysoberyl | This semi-transparent oval cabochon, coloured a typical deep yellow known as milk and honey, features a finely formed cat's-eye pattern.



Cat's-eye oval | This oval cabochon of yellow-green cat's-eye chrysoberyl has an unusual bluish cast around the area of the "eye", and blemishes on its surface.

Cut



Alexandrite crystals | This specimen of alexandrite with mica was mined in Siberia, Russia, which remains a major source of high-quality material.



Twinned crystals | These two crystals growing into a "V" formation from the same point at their base show classic chrysoberyl twinning.



Cat's-eye cabochon | This transparent, cabochon shows not only the distinctive "eye" pattern but also the fibrous inclusions that created it, all in a hazy yellowish hue.



Green brilliant cut | The pale green colour of this stone gives it a luminous shine, emphasized by its unusual 10 main facets, rather than the eight on a standard brilliant.